MIL-P-48296(PA)
1 May 1974
SUPERSESSION
(see 6.3)

MILITARY SPECIFICATION

POLYTETRAFLUOROETHYLENE (TFE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope.This specification covers one type of polytetrafluoroethylene for use in pyrotechnics (see 6.4).

1.2 Classification. The TFE shall be classified as follows:

Class 1 - Particle size, 95 microns
Class 2 - Particle size, 237 microns
Class 3 - Particle size, 200 microns

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1233- Procedures for Determining Particle Size, Particle Size Distribution and Packed Density of Powdered Material

FSC: 9330

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MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, drawings and publications required by suppliers in connection with specific procurement functions shall be obtained from the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation E168 - General Techniques of Infrared Quantitative Analysis, Standard Recommended Practices for

ASTM Designation E300 - Industrial Chemicals, Recommended Practice for Sampling

ASTM Designation D1457 - Tetrafluoroethylene Resin Molding and Extrusion Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Penn. 19103).

3. REQUIREMENTS

3.1 Material. The material furnished under the specification shall be polytetrafluoroethylene (TFE), conforming to the following requirements.

3.1.1 Composition. The purity of TFE shall be 99.4% minimum when determined as specified in 4.4.1.

3.1.2 Infrared spectrum. The TFE shall have an infrared spectrum with peaks consistent with those in Figure 1 when tested as specified in 4.4.2. The infrared spectrum shall contain no extraneous significant peaks. A significant peak is defined as a peak having an absorbance greater than 10 percent beyond the normal background.

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3.1.3 Color. The TFE shall be opaque and the color shall range from white to gray, when determined as specified in 4.4.3. The gray shall be no darker than Chip No. 36559 of Table VIII in FED-STD-595.

3.1.4 Moisture. The moisture of the TFE shall be 0.05 percent maximum, when determined as specified in 4.4.4.

3.1.5 Ash. The ash of the TFE shall be 0.1 percent maximum, when determined as specified in 4.4.5.

3.1.6 Melting point. The melting point of the TFE shall be 327 degrees centigrade(°F) ± 10°C when determined as specified in 4.4.6.

3.1.7 Packing density. The packing density of TFE, when determined as specified in 4.4.7, shall be:

Class 1 - 1.18 ± 0.13 gm/ml
Class 2 - 1.25 ± 0.02 gm/ml
Class 3 - 1.14 ± 0.09 gm/ml

3.1.8 Particle size. The particle size of the TFE when determined as specified in 4.4.8, shall be:

Class 1 - 95 ± 15 microns
Class 2 - 237 ± 27 microns
Class 3 - 200 ± 30 microns

3.1.9 Particle size distribution. The particle size distribution shall be as specified in Table I when determined as specified in 4.4.9.

TABLE I

Sieve Size	Percent Retained On		
	Class 1	Class 2	Class 3
40	-	None	-
50	-	13-24	0-6
60	0.8 max	15-35	2-18
70	-	10-22	-
80	-	8-14	38-57
100	5 max	7-16	-
120	11-29	-	20-37
140	15-26	-	-
170	11-21	-	-
200	7-16	7-16	1-13
325	9-25	-	-
Pan	9-26	6-12	6 max.

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3.1.10 Particle shape. The particle shape of the TFE shall be as indicated in Figure 2. The TFE shall be free from hair shape or needle shape particles when determined as specified in 4.4.10.

3.2 First article testing. This specification makes provisions for first article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein.

4.1.1 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Specification number and date, together with identification and date of changes thereto.

c. Certificates of analysis on all materials used directly by the contractor when such material is controlled by Government specifications shall be made available upon request by the Contracting Officer.

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d. Quantity of TFE in pounds in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission. Prior to the start of regular production the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. The first article shall consist of ten (10) pounds (lbs) of TFE in accordance with instructions issued by the Contracting Officer. All samples submitted shall have been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.3 and 4.4 of this specification.

4.2.3 Rejection. If any sample fails to comply with any of the applicable requirements, the first article

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quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation. A lot shall consist of one or more batches of polytetrafluoroethylene produced by one manufacturer, in accordance with the same specification, or same specification revision under one continuous set of operating conditions. Each lot shall consist of that quantity of polytetrafluoroethylene that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordnace with MIL-STD-105. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.2 Examination. Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105), except that inspection for critical defects shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect, except where 100 percent inspection is specified.

4.3.2.1 Containers selected

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined.			
Major: None defined.			
Minor:	AQL 1.00 percent		
201.	Container damaged so that contents are exposed or liable to become exposed	Visual	01001

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202. Marking missing, incorrect or illegible	Visual	01002

4.3.3 Sampling. The tests depicted in 4.4.1 through 4.4.10 shall be performed on a representative sample from each batch. Approximately 500 grams of TFE shall be withdrawn from each batch to be sampled using ASTM Procedure E300 for solids. Samples shall be selected for inspection in accordance with MIL-STD-1235, CSP-1 Plan, Inspection Level II, AQL 6.5%. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected. All batches produced between the time that the last batch was tested and accepted and the batch which failed shall be tested in accordance with the applicable methods given in paragraph 4.4. If any of these batches fail to meet any of the test requirements, that batch shall also be rejected. In addition, after any failure of a batch the contractor will return to 100% inspection until "i" successive batches are accepted as required by MIL-STD-1235. The classification and code number shall be given in Table II.

TABLE II

TFE content(see 3.1)	Major defect	02001
Infrared Spectrum(see 3.1.2)	Major defect	03001
Color (see 3.1.3)	Major defect	04001
Moisture (see 3.1.4)	Major defect	05001
Ash (see 3.1.5)	Major defect	06001
Melting point(see 3.1.6)	Major defect	07001
Packing density (see 3.1.7)	Major defect	08001
Particle shape(see 3.1.8)	Major defect	09001
Particle size distribution(see 3.1.9)	Major defect	10001
Particle shape (see 3.1.10)	Major defect	11001

4.3.4 Inspection Equipment. The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the

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required accuracy. Commercial inspection equipment, shall be employed where applicable for all tests and examinations specified in 4.4. The contractor is responsible for assuring proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purpose.

4.4 Test methods

4.4.1 Polytetrafluoroethylene (TFE) content. The TFE content shall be determined as follows:

$$\% \text{ TFE} = 100 - (A + B)$$

Where:
A = % moisture, determined in 4.4.4.
B = % ash, determined in 4.4.5.

4.4.2 Infrared spectrum. Determine the infrared spectrum of the sample of material using an Infrared Spectrophotometer. Preparation of sample and method used shall be in accordance with the instructions in ASTM-E-168.

4.4.3 Color. Take a portion of approximately 50 grams from the sample and spread the examination sample over a flat surface. Examine the material for color using daylight illumination to determine compliance with 3.1.3.

4.4.4 Moisture. The moisture content shall be determined in accordance with ASTM D1457.

4.4.5 Ash. Accurately weigh a sample of approximately 10 grams and place it in a previously ignited tared platinum crucible. Ignite the crucible and contents in a muffle furnace maintained at 650°C for thirty minutes, cool for one hour in a desiccator and weigh. Repeat the ignition, cooling and weighing until the change in weight between weighings does not exceed 0.2 mg. Calculate the ash content as follows:

$$\text{Percent ash} = \frac{A}{W} \times 100$$

Where:
A = weight of ignited sample, grams
W = weight of sample, grams

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4.4.6 Melting point. The melting point shall be determined in accordance with ASTM D1457.

4.4.7 Packing density. The packing density shall be determined in accordance with MIL-STD-1233, Method 400.

4.4.8 Particle size

4.4.8.1 Class 1. The particle size shall be determined in accordance with ASTM D1457, except the sieve size shall be 60, 70, 80, 100, 120, 140, 170, 200 and 325. The 50 percent point shall be determined by drawing the best straight line through 120, 140, 170 and 200.

4.4.8.2 Class 2. The particle size shall be determined in accordance with ASTM D1457, except that sieve size shall be 40, 50, 60, 70, 80, 100, 200 and pan. The 50 percent point shall be determined by drawing the best straight line through 50, 60, 70, 80 and 100.

4.4.8.3 Class 3, The particle size shall be determined in accordance with ASTM D1457, except that sieve size shall be 50, 60, 70, 80, 100, 120, 140, 170, 200 and pan. The 50 percent point shall be determined by drawing the best straight line through 50, 60, 80, 120 and 200.

4.4.9 Particle size distribution. The particle size distribution shall be determined as specified in 4.4.7, except the percent retained and the percent through the required sieve shall be determined.

4.4.10 Particle shape. Spread a thin layer of the sample on a glass slide and examine under a microscope using a magnification of approximately 100.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing shall be as specified in the contract or purchase order.

5.2 Marking. Marking shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this document.

b. Acceptance and description sheets. Acceptance and description sheets shall be prepared for each lot in accordance with MIL-STD-1171.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.3 Supersession Data. This specification includes the requirements of Purchase Description PA-PD-2491, dated 8 May 1963.

6.4 Intended use. The material in this specification is intended for use for application in pyrotechnic and flare compositions (i.e. MX 3529)

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Army-PA

Preparing Activity:
Army-PA

Project Number: 9330-A661

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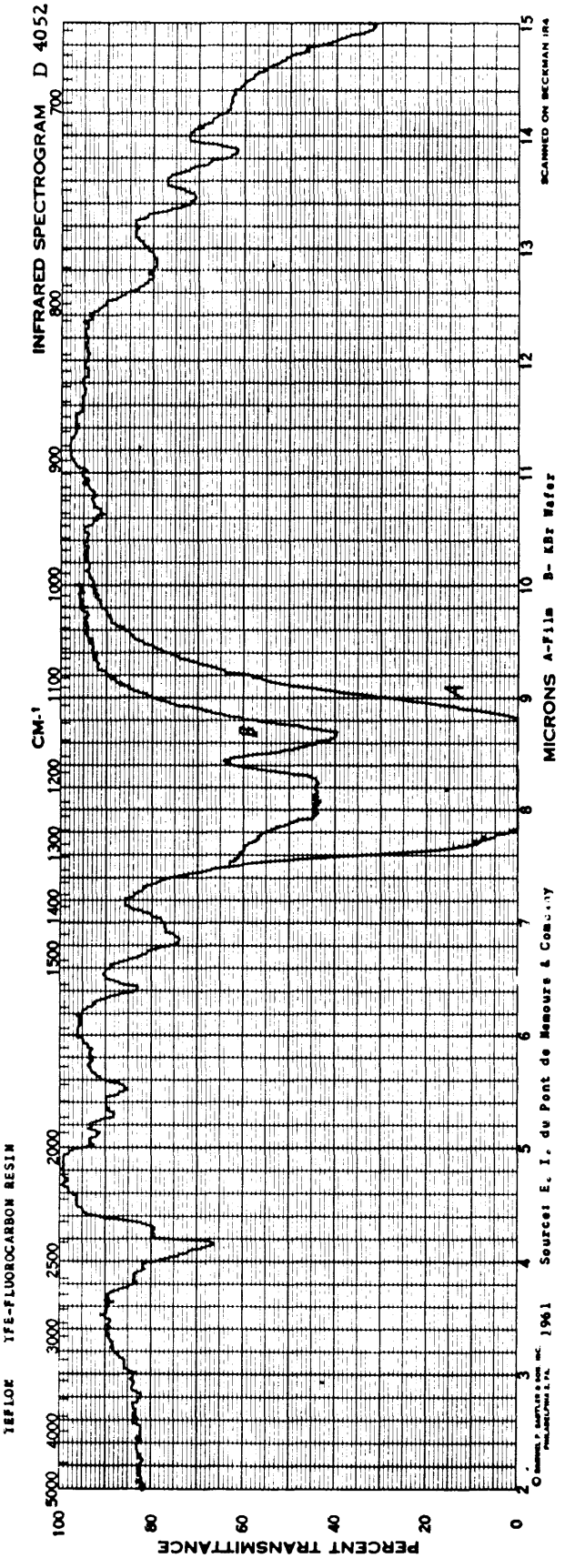


FIGURE 1

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Particle Shape of Polytetrafluoroethylene Resin

Figure 2

100 Magnification

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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